

XIN FENG

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Research Areas: Computer Vision, Generative Models, Image Processing,
Representation Learning, Lifelong Learning, 3D Content Generation, Vision-Language Models

EDUCATION

- Ph.D. (Computer Science and Technology)** Sept. 2019 – Aug. 2023
Harbin Institute of Technology, Shenzhen, China
Supervisors: Prof. Guangming Lu, Prof. Wenjie Pei
Thesis: Research on Degradation Pattern-Aware General Image Restoration
Contributed to the development of general image restoration from three perspectives: architecture, generative priors, and generalization.
- Proposed a unified generative framework for background restoration from superimposed images (IEEE TIP 2021).
 - Developed a memory-guided semantic reasoning model and a nested U-shaped Transformer for general image restoration (IEEE TCSVT 2022&2023).
 - Proposed hierarchical contrastive learning for pattern-generalizable image corruption detection (ICCV 2023).
- M.Eng. (Control Engineering)** Sept. 2017 – June 2019
Beijing Forestry University, Beijing, China
Supervisor: Prof. Jiangming Kan
Thesis: Pseudo Entropy-based Architectural Optimization Algorithm for Deep Neural Networks
Research: Neural Network Pruning, Depth Estimation
- B.Eng. (Computer Science) & B.A. (English)** Sept. 2013 – June 2017
Zhengzhou University, Zhengzhou, China
Supervisors: A/Prof. Changyong Niu (CS), A/Prof. Yumin Zhao (English)
Thesis: Collaborative Filtering-based Music Recommendation Algorithm
Research: Machine Learning, Recommendation Algorithms, Data Mining

EXPERIENCE

- Research Associate, University of Edinburgh, U.K.** Apr. 2025 – Present
CoSTAR Real-time Lab, School of Informatics.
- Developed the **first** framework for 3D human motion style transfer with morphology control.
 - Researched anomaly detection and correction methods for motion-capture (MoCap) data.
 - Currently developing novel methods for 3D asset generation.
- Technical Section Chief, China Tobacco** Sept. 2023 – Mar. 2025
Information Center, applied AI research and deployment in industrial manufacturing.
- Applied for and led a real-time cross-modality anomaly detection project for manufacturing-line monitoring.
 - Applied for and led the development of an LLM-based intelligent knowledge base for manufacturing.
 - Delivered AI-based systems spanning large language models, anomaly detection and video understanding; filed 3 patents (currently confidential).
- Low-level Vision Research Intern, Huawei Noah's Ark Lab, Shenzhen** June 2022 – Sept. 2022
Supervisor: Dr. Xueyi Zou.
- Developed an image super-resolution algorithm utilizing pre-trained generative models to enhance visual quality.
- Video Understanding Research Intern, iQIYI Technology, Beijing** Aug. 2018 – June 2019
Supervisor: Dr. Ruijin Jin.
- Developed a non-local 3D convolution-based video understanding model to score entertainment value of film and TV products.
 - Developed a GAN-based algorithm for image harmonization to integrate advertising content with television content.
- Visiting Intern, Peking University, Beijing** June 2018 – Aug. 2018
- Investigated improving password-guessing performance using GAN models.

PUBLICATIONS (Google Scholar)

Journal	Learning Sequence Representations by Non-local Recurrent Neural Memory Wenjie Pei ¹ , Xin Feng ¹ , Canmiao Fu, Qiong Cao, Guangming Lu [†] , Yu-Wing Tai. <i>International Journal of Computer Vision (IJCV)</i> , 2022. (IF: 11.6)
	Deep-Masking Generative Network: A Unified Framework for Background Restoration from Superimposed Images Xin Feng , Wenjie Pei, Zihui Jia, Fanglin Chen, Guangming Lu [†] , David Zhang. <i>IEEE TIP</i> , 2021. (IF: 10.8)
	Generative Memory-Guided Semantic Reasoning Model for Image Inpainting Xin Feng , Wenjie Pei, Fengjun Li, David Zhang, Guangming Lu [†] . <i>IEEE TCSVT</i> , 2022. (IF: 8.3)
	U²-Former: Nested U-shaped Transformer for Image Restoration via Multi-view Contrastive Learning Xin Feng , Haobo Ji, Wenjie Pei, Jinxing Li, Guangming Lu [†] , David Zhang. <i>IEEE TCSVT</i> , 2023. (IF: 8.3)
	GAN Inversion-based Semi-supervised Learning for Medical Image Segmentation Xin Feng , Jianyong Lin, Chun-Mei Feng, Guangming Lu [†] . <i>Biomedical Signal Processing and Control (BSPC)</i> , 2024. (IF: 4.9)
Conference	Hierarchical Contrastive Learning for Pattern-Generalizable Image Corruption Detection Xin Feng , Yifeng Xu, Wenjie Pei, Guangming Lu [†] . <i>ICCV</i> , 2023.
	Learning Generalizable Latent Representations for Novel Degradations in Super-Resolution Fengjun Li ¹ , Xin Feng ¹ , Fanglin Chen, Guangming Lu, Wenjie Pei [†] . <i>ACM MM</i> , 2022.
	Contrastive Feature Decomposition for Image Reflection Removal Xin Feng , Haobo Ji, Bo Jiang, Wenjie Pei, Fanglin Chen, Guangming Lu [†] . <i>ICME</i> , 2021. (Oral)
Preprint	Global-local Stepwise Generative Network for Ultra High-resolution Image Restoration Xin Feng , Haobo Ji, Wenjie Pei, Fanglin Chen, Guangming Lu [†] . <i>arXiv</i> , 2022.
Submitted	Restore without Forgetting: Filter-level Continual Adaptation via Integrated Gradient Attribution Xin Feng , Jin Zhao, Yizhen Zhang, Wenjie Pei, Fanglin Chen, Guangming Lu [†] . 2026.
	MorphoStyle: Motion Style Transfer with Morphology Control Xin Feng , Eleonora D'Arnese, Mohan Sridharan [†] . 2026.

AWARDS AND ACHIEVEMENTS

Outstanding Doctoral Dissertation Award, Shenzhen Association for Artificial Intelligence	2024
National Scholarship for Ph.D. Students (top 0.2%), Ministry of Education, China	2022

TEACHING AND SUPERVISION

Guest Lecturer, University of Edinburgh, U.K.	2026
• INFR11132 Machine Learning Practical (MSc.): delivered a guest lecture on Representation Learning and Generative AI.	
Teaching Assistant, Harbin Institute of Technology, Shenzhen, China	2021
• Teaching assistant for Digital Image Processing : supported lectures and tutorial sessions.	
Co-supervised Students	
University of Edinburgh	
Erastus Rebiro (with Prof. Frauke Zeller);	
Kanishka Roy (with Dr. Theodore Koterwas).	
Harbin Institute of Technology	
Yifeng Xu, Yizhen Zhang, Hongyu Xia (with Prof. Wenjie Pei);	
Haobo Ji (with Prof. Guangming Lu);	
Jin Zhao (with A/Prof. Fanglin Chen).	

TALKS AND PRESENTATIONS

BMVA Symposium on Continual AI, British Computer Society, London.	2026
<i>Restore without Forgetting: Filter-level Continual Adaptation via Integrated Gradient Attribution.</i>	

SELECTED PROJECTS

Palm Vein Image Reflection Restoration and Recognition, Tencent (participant)	2023
Video Understanding based on Deep Semantic Spatio-Temporal Representation, NSFC (participant)	2022
Research on Structural Optimization by Dynamic Perception for Deep Models, NSFC (participant)	2021

ACADEMIC SERVICE

Workshop Committee

Committee Member, [Physical Principles for Reliable 3D Modelling in Multimedia \(P3DMM\)](#), IEEE ICME 2026 Workshop. 2026

Conference Reviewer

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
 IEEE/CVF International Conference on Computer Vision (ICCV)
 European Conference on Computer Vision (ECCV)
 ACM Special Interest Group on Computer Graphics and Interactive Techniques (ACM SIGGRAPH)
 Conference on Neural Information Processing Systems (NeurIPS)
 AAAI Conference on Artificial Intelligence (AAAI)
 ACM International Conference on Multimedia (ACM MM)
 British Machine Vision Conference (BMVC)
 Asian Conference on Computer Vision (ACCV)

Journal Reviewer

IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE T-PAMI)
 IEEE Transactions on Image Processing (IEEE TIP)
 IEEE Transactions on Multimedia (IEEE TMM)
 IEEE Transactions on Neural Networks and Learning Systems (IEEE TNNLS)
 IEEE Transactions on Circuits and Systems for Video Technology (IEEE TCSVT)
 IEEE Transactions on Systems, Man, and Cybernetics: Systems (IEEE TSMC)
 Pattern Recognition (PR)
 Computer Vision and Image Understanding (CVIU)
 Applied Soft Computing (ASC)
 Knowledge-Based Systems (KBS)
 Biomedical Signal Processing and Control (BSPC)
 Artificial Intelligence in Medicine (AIIM)